

# Wireless EMG Controlled PROSTHETIC HAND

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**W**hat if you had an extra pair of hands to carry out tasks somewhere else while you controlled them from another location wirelessly. Seems unrealistic, right? But it's possible.

This project describes how to design a prosthetic hand that can replicate natural hand movements and muscle contractions. Although artificial, it works like a biological hand by following the muscle signals.

To achieve this, the muscle contraction and stress values are captured and then through muscle signals used to wirelessly control the

prosthetic arm located somewhere else. The project can be developed further to help the physically challenged and also to grip objects lying in places where you do not want to dirty your own hands.

## Assembly

There are various open source prosthetic body parts available in the market. You can also 3D print them at home. In this project the InMoov robotic arm has been used. After getting the required parts comprising fingers and

servo, assemble the prosthetic hand as shown in the figures.

## Coding

We need to make two devices—a

**TABLE 1**  
**MATERIALS**

| Component name             | Quantity | Description          |
|----------------------------|----------|----------------------|
| 3D Printed prosthetic hand | 3        | Prosthetic hand      |
| Arduino Pro Micro          | 2        | ATmega32u4           |
| Jumper wires               | 6        | Female to female     |
| 9V battery                 | 2        | 9V DC MN1604 battery |
| Servo motor                | 1        | 12kg/cm <sup>2</sup> |
| EMG sensor                 | 1        | EMG sensor           |
| Bluetooth HC 05            | 2        | Bluetooth module     |



Fig. 1: Inside view of the prosthetic arm



Fig. 3: Outside view of the prosthetic arm



Fig. 2: The prosthetic arm with servo move mechanism



Fig. 4: The prosthetic hand